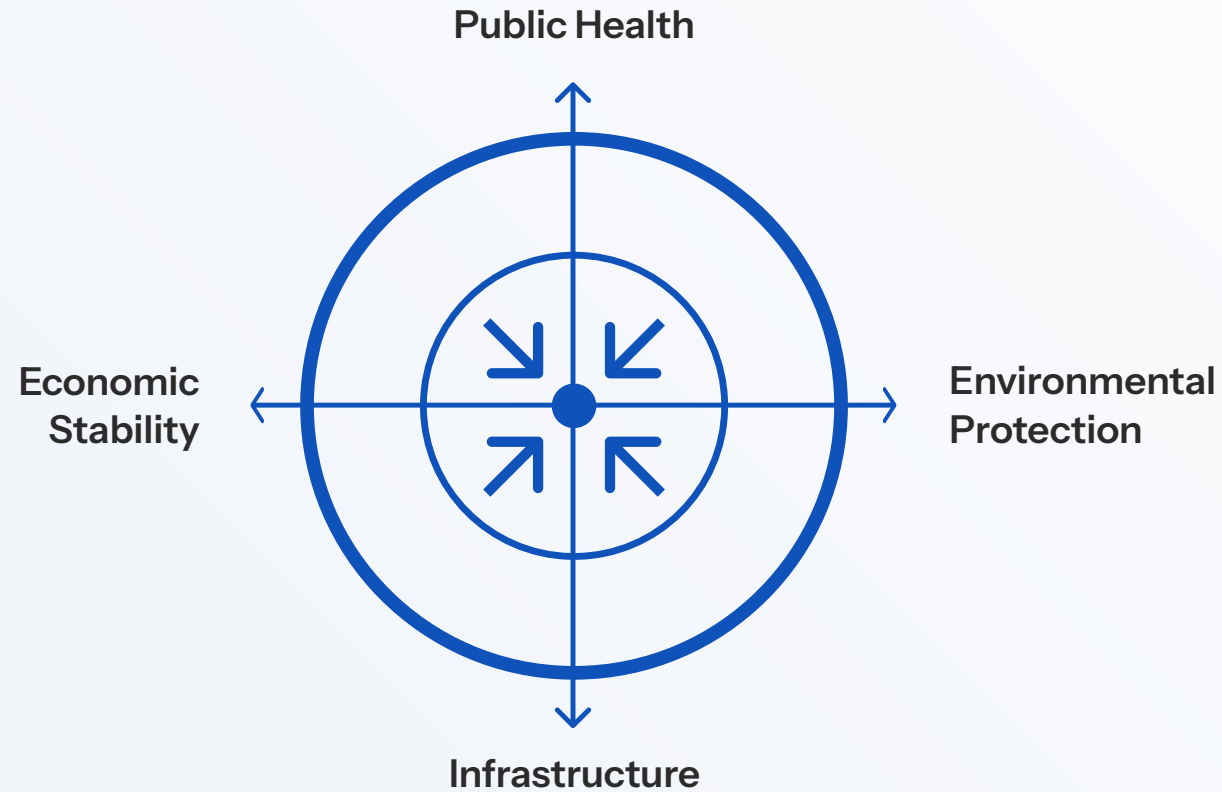


Water Treatment Optimization System

Chemical Engineering • Data Science • Process Optimization • Sustainability • AI. Predicting water quality, optimizing chemical dosing, and estimating energy usage through intelligent, data-driven engineering.



Introduction — Why Do We Care About Water Treatment?



Water treatment systems protect public health by removing bacteria, heavy metals, suspended solids, organic pollutants, excess nutrients, and industrial contaminants. Without effective treatment, communities face disease outbreaks, unsafe drinking water, and unsustainable agricultural and manufacturing conditions. **Water treatment is not only an environmental issue — it is a public health, engineering, economic, and national infrastructure issue.**

Why This Project Matters

Traditional water treatment relies on fixed chemical dosing rules and manual operator decisions. AI allows treatment plants to become smarter — shifting from reacting to poor water quality to **predicting and preventing it** before it occurs.

This project builds a data-driven system connecting environmental engineering, chemical engineering, public utilities, smart cities, and AI-driven infrastructure.



What the System Does & What It Measures

Objectives

01

Predict Final Water Quality

Forecast treated turbidity, pH, and compliance pass/fail status before water leaves the plant

02

Recommend Optimal Chemical Dosage

Reduce chemical waste while meeting all regulatory standards for chlorine and coagulants

03

Estimate Energy Consumption

Model pump power and aeration energy usage to minimize operational costs

04

Detect Abnormal Conditions

Flag contamination events and process anomalies before they cause compliance failures

05

Support Sustainable Infrastructure

Improve regulatory compliance, reduce waste, and support long-term environmental goals

Input Variables

Raw Water

Turbidity, pH, temperature, flow rate, total dissolved solids

Process

Chemical dosage, chlorine residual, pump pressure, aeration rate, energy consumption

Output Variables

Quality

Treated turbidity, final pH, compliance pass/fail

Cost

Chemical cost, energy usage, treatment efficiency

Key Formulas & Engineering Principles

Mass Balance (Steady-State)

$$Input = Output$$

Tracks contaminants, chemicals, and flow through the treatment system.

Concentration

$$C = \frac{m}{V}$$

Where C = concentration, m = mass of contaminant, V = volume of water.

Removal Efficiency

$$Removal\ Efficiency = \frac{C_{in} - C_{out}}{C_{in}} \times 100$$

Measures how effectively a treatment stage removes a given contaminant.

Chemical Dosing

$$Chemical\ Mass = Dose \times Flow$$

Determines daily chemical requirements in mg/L, ppm, or kg/day.

Hydraulic Retention Time

$$HRT = \frac{V}{Q}$$

Where V = tank volume, Q = flow rate. Determines how long water stays in a treatment tank.

Pump Power

$$P = \frac{\rho g Q H}{\eta}$$

Where ρ = water density, g = gravity, Q = flow rate, H = pump head, η = efficiency.

Energy Usage

$$Energy = Power \times Time$$

Measured in kWh. Used to model operational cost for pumps and aeration systems.

Optimization Objective

$$Minimize : Cost = Chemical\ Cost + Energy\ Cost + Penalty\ Cost$$

Subject to: turbidity $\leq$ standard; pH and chlorine within permitted bounds.

System Design & Workflow



Collect & Preprocess Data

Gather raw water quality readings from sensors, flow meters, pH meters, and turbidity analyzers. Clean and normalize data for modeling.



Predict Quality & Dosage

Apply ML models to forecast treated water turbidity, recommend chemical dosage, and estimate energy usage for current inlet conditions.



Compare Against Regulatory Limits

Automatically evaluate predicted outputs against turbidity standards, pH bounds, and chlorine residual compliance thresholds.



Recommend Settings & Visualize KPIs

Output optimized operating parameters and display real-time KPI dashboards for plant operators and compliance teams.

Technologies Used & Machine Learning Models

Technology Stack

Programming & Analytics

Python 🐍, Pandas, NumPy, Scikit-Learn, Matplotlib, SQL, Excel

Visualization & Reporting

Power BI, Jupyter Notebook, optimization models

Operations & Sensors

SCADA, PLC data, flow meters, pH meters, turbidity meters, chlorine analyzers

ML Models & Prediction Targets

Regression Models

Linear Regression, Random Forest, Gradient Boosting, XGBoost, Neural Networks — predicting turbidity, dosage, and energy

Anomaly Detection

Isolation Forest — flags process anomalies, contamination events, and compliance risk in real time

- Final turbidity & chemical dosage required
- Energy usage & compliance risk score
- Process anomaly probability

Implementation Example: Water Quality Prediction Model

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error

# Load dataset
df = pd.read_csv("water_treatment_data.csv")

features = [
    "raw_turbidity", "raw_ph", "temperature",
    "flow_rate", "chemical_dose", "pump_pressure"
]
target = "treated_turbidity"

X = df[features]
y = df[target]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

model = RandomForestRegressor(random_state=42)
model.fit(X_train, y_train)

predictions = model.predict(X_test)
mae = mean_absolute_error(y_test, predictions)

print("Water Quality Prediction Model Complete")
print(f"Mean Absolute Error: {mae:.3f}")
```

What This Code Does

1 Loads sensor data

Reads raw turbidity, pH, temperature, flow rate, chemical dose, and pump pressure from a CSV dataset

2 Trains a Random Forest

Fits a regression model to predict treated water turbidity from inlet process conditions

3 Evaluates accuracy

Measures Mean Absolute Error on held-out test data to validate prediction performance

Key Performance Indicators & Industry Applications

Key Performance Indicators

KPI	Meaning
Treated Water Turbidity	Clarity of treated water
Final pH	Chemical balance
Removal Efficiency	Treatment performance
Chemical Usage	Cost and dosage control
Energy Usage	Operational efficiency
Compliance Rate	Regulatory performance
Cost per Gallon Treated	Economic performance
Pump Efficiency	Energy optimization

Industry Applications



Municipal Water Treatment

Cities providing safe drinking water to residents



Industrial Wastewater

Factories, refineries, chemical plants, and manufacturing sites



Power Plants & Data Centers

Cooling water, boiler feedwater, and sustainability monitoring



Agriculture & Healthcare

Irrigation water quality, runoff control, and purified water systems

Optimization: Before vs. After

The Measurable Difference Data-Driven Treatment Makes

Without optimization, treatment plants operate reactively — wasting chemicals, consuming excess energy, and risking compliance failures. With water treatment optimization, every parameter is managed proactively.



Excessive Chemical Use → Reduced Waste

Dosing models eliminate over-treatment, cutting chemical costs without sacrificing quality



High Energy Costs → Lower Consumption

Pump power optimization and aeration modeling reduce operational electricity usage



Regulatory Violations → Compliance Assurance

Predictive models flag compliance risk before treated water leaves the plant



Public Health Risks → Safer Communities

Consistent, high-quality treated water protects residents and supports city resilience

Skills Demonstrated & Future Improvements

Skills Demonstrated

Chemical Engineering

Mass balance, fluid flow, chemical dosing, reaction kinetics, process control

Data Science

Regression modeling, forecasting, anomaly detection, KPI dashboards, predictive analytics

Environmental Engineering

Water quality monitoring, contaminant removal, treatment efficiency, compliance analysis

Business Analytics

Cost reduction, energy optimization, operational efficiency, sustainability reporting

Future Improvements

01

Real-Time SCADA Dashboard

Live monitoring of all process variables and KPIs

02

AI Chemical Dosing Engine

Automated recommendation system for coagulant and chlorine dosing

03

Digital Twin

Full simulation model of a water treatment plant for scenario planning

04

Predictive Maintenance

Anomaly detection for pumps and early warning for contamination events

05

Power BI Sustainability Dashboard

Executive-level compliance and environmental reporting

Final Statement

This project demonstrates how chemical engineering, environmental science, data analytics, and artificial intelligence can work together to improve water quality, reduce chemical waste, lower energy usage, and protect public health.

In the AI era, water treatment optimization is not only an engineering project — it is a **society-level solution for sustainability, infrastructure resilience, and human well-being**. Better water treatment means healthier people, cleaner environments, and stronger infrastructure.

